



VD – Tire Training

Nicolas Thirion

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Introduction





Scope

The objective of this training module is to provide the trainee with a deep understanding for tire fitting and modeling.

A strong base in MATLAB is recommended to complete this training. If you've never used MATLAB before, is strongly advised that you complete their [MATLAB on-ramp course](#) before continuing.

You will be using GitHub to access scripts and save/share your progress with others. This tool is used by the entire team and base knowledge of how to use it is required. If you've never used GitHub, follow this [introductory training](#). Only the first few steps are necessary to get yourself started but refer to this as documentation for the future.

Software	Version
MATLAB	R2022a
GitHub	3.0.7



Setup

Using Github, you will need to clone the '*Tyre_model*' repository to your computer. Then create a branch named : '*Training_YourInitials*'. This will serve as your working folder. When working in this branch, feel free to delete/modify/add as you wish. Any changes you make should be committed with relevant naming and a short description of the changes.

Do NOT push your changes to the main branch. Ensure you are in your branch before pushing any changes.

Dig through the scripts in the Tyre_model working folder. Get familiar with the content and how it all interacts together, we will be moving forward assuming you know where to find referenced scripts.



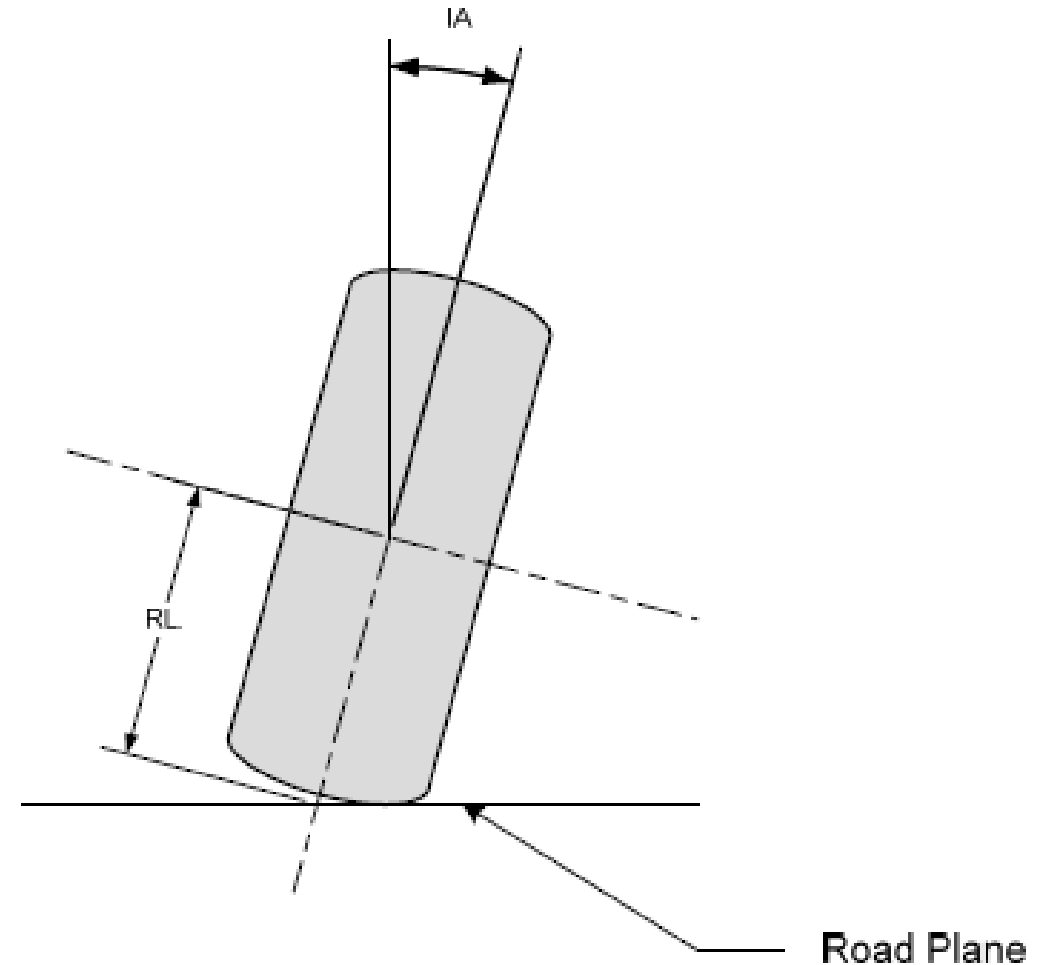
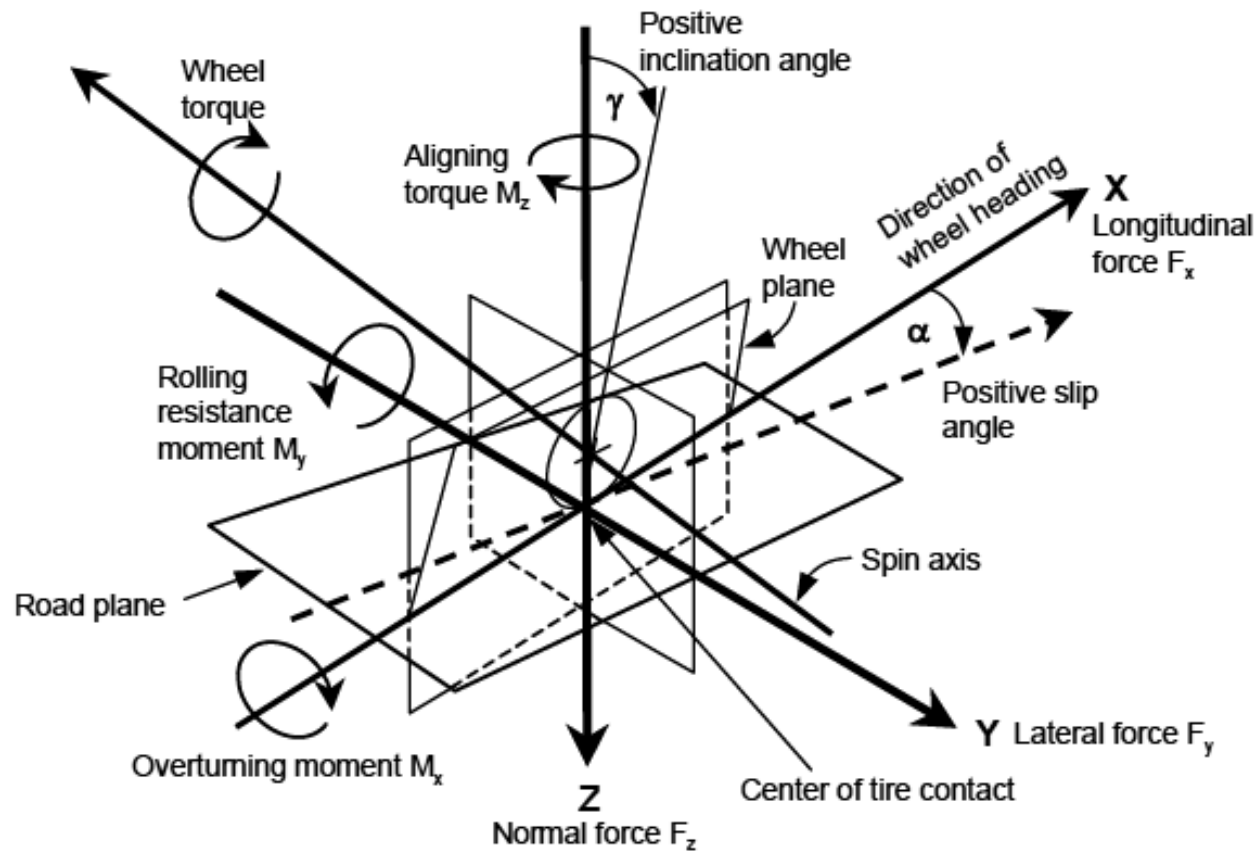
Tire Forces and Moments

The base of any tire analysis is to understand what are the different forces and moments acting on the tire and how do they impact the driver.

Here is a [short video](#) explaining tire moments and forces.

Tire Axis Definitions

$$\text{Torque } T = F_x R_l + M_y \cos \gamma + M_z \sin \gamma$$



Tire Data



Data Collection Background

As an FSAE team, we have access to an extraordinary amount of tire data. Our team has paid a 500\$ entry fee to the TTC (*Tire Testing Consortium*) to have access to tire data tested by Calspan.

The FSAE TTC website has a forum which is a gold mine of questions for data processing, analysis, tire selection and anything related to tires really. There is never just one right answer to solving these tire problems, reading through this forum is great to spark new ideas and approaches to various problems.

Here is a [short video](#) demonstrating a tire test rig in action.

Accessing The Data

To access the data, you will need to create an account using your McGill details on the [TTC website](#). It will take a day or two for your account to be approved. This is a good time to review some of the tire modeling scripts! Once you have access, you will need to download the [data from the TTC website](#).

You will want to download two zip files:

- Download '*fromCalspan_Round9.zip*' which contains:
 - RunGuide_Round9 : A table containing all tires, their run number (or file number) for each test and a short description of each test
 - Contents_Round9 : A thorough description of the tire tests and the data channels. **MUST READ THROUGH THIS.**
 - And few more documents that you should investigate which will help once we start processing the data.
- Download '*RawData_Cornering_Matlab_SI_Round9_Runs1to15*'
 - This folder contains test data for runs 1 to 15 in SI units and '*.mat*' format.
 - Refer to '*RunGuide_Round9*' to understand which tires were tested for these runs and take note of any events that might have affected the data.

Data Preprocessing

If you plot the raw data in MATLAB of any run and zoom in to a small section, you will notice there is a large amount of noise. Before using the data for a fit, you will need to preprocess it. Our current process follows 4 major steps. Some data sets will require extra processing such as chopping and concatenating. One of the particularities you will notice for one of the tires is that a full test is separated over two data files due to the pressure hose detaching from the rim.

Preprocessing steps:

1. Cropping : Segmenting the data sets into the different pressures.
2. Smoothing : Filtering each segment with a .1 Hz low-pass filter at a 50 Hz sampling frequency.
3. Grouping : Creates combined and cornering files that combines all the pressures.
4. Restructuring : Creates a .mat containing the all the data in a structured format (per normal loads, camber, pressures)

To improve the fit, the cropping step should be completed diligently, and the filtering frequency should be adjusted for each data set. When the final processed is plotted, the curves should be as clearly defined as possible.

Model Fitting



Equations

There are many tire models out there... From simple models that only require a handful of coefficients, to very complex models that include tire temperature models and become very representative of the actual tire. The more complex the model, the more test data is required.

For our team's purposes, we only model the force-moment handling characteristics. We also run a DIL which requires a more complex model which is not discussed here. The force-moment equations are used in steady-state analysis such as Ackermann, yaw moment diagram, tire selection, kinematics target selection, etc. Our MATLAB script currently use the MF6.1 force-moment equations which allow us to also model how the tire pressure impacts performance.

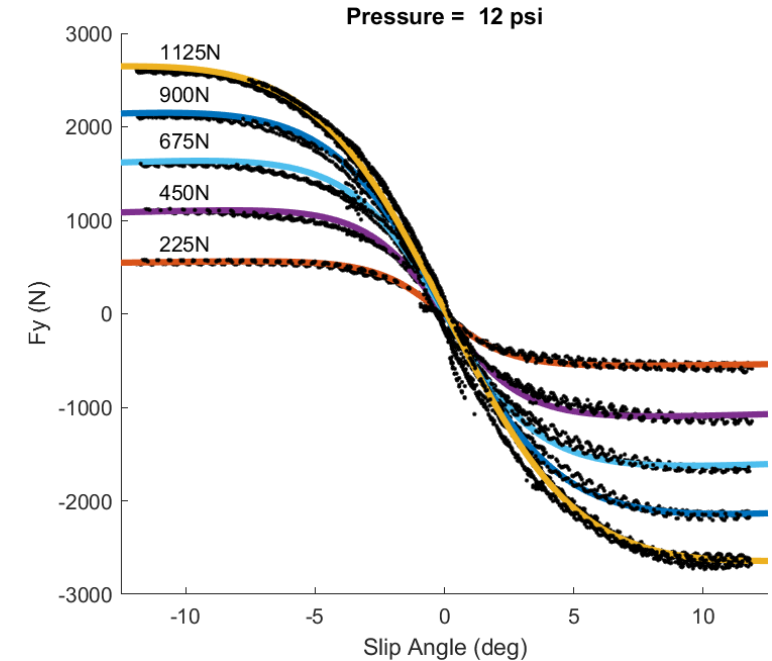
The equations were taken from '*MF-Tyre & MF-Swift 6.2 Equation Manual.pdf*' which can be found under '*7 - MF-Tyre & MF-Swift Model Reference*'.

- Pay attention to the various sections and to the difference between coefficients and scaling factors.

Fitting

To fit a tire model using the already created scripts, go to:

- 3 - Fitting Magic Formula Parameters
 - Magic Formula 6.1
 - 1 - Lateral Force Pure Slip Fitting
 - Ensure this is your working folder



You can now run the fitting script, select your tire, and let the magic begin. Once the fit is complete, you will need to compare this back to data. To do so, change your working folder to '*Magic Formula 6.1*' and open the script '*Evaluate_Fitting.m*'. This script will overlay the cleaned-up data and the fit for various tire conditions. Once completed, proceed to the next fit section and repeat this process to the end.

Script Setup

To generate your own script, you will need to make use of the pure slip and combined data sets separately. Most of the model sections contain variables and coefficients that are found in the pure slip sections. For this reason, the model coefficients are fitted in a particular order:

1. Lateral Pure Slip
2. Lateral Combined
3. Longitudinal Pure Slip
4. Longitudinal Combined
5. Self-Aligning Moment in Pure Lateral
6. Self-Aligning Moment Combined
7. Overturning Moment Combined
8. Rolling Resistance Moment

Cost Function

To write a script that automatically fits the model coefficients to the data you feed it, you will need to setup cost functions to use within your optimization script. A cost function will evaluate your root mean squared error (RMSE) at each iteration. This error is calculated between each data point given and the fit.

In our case, using lateral pure slip as an example, when executing the optimization function, we call '*RMSE_Lateral_PureSlip_Step1.m*' which will run the given coefficients through the model equations and evaluate the RMSE. This plays the roll of stopping criterion for the optimization script.

Optimization

The optimization function serves the purpose of changing the model coefficients until the RMSE falls under a given value. At each fitting step, only the essential parameters for the equations are sent to the optimization script to be modified. A starting value, an upper bound and a lower bound are also provided to guide the optimization script and reduce computation time. The coefficients bounds can impact the results and should be tuned accordingly for each new tire.

In our scripts, most fitting sections are completed over multiple optimization steps. The objective of the steps is to provide the minimum amount of data to the optimization to help it narrow down on the solution slowly. Providing all the data at once will make computation lengthy and often result in a bad fit. This improves the process's robustness.

There are many optimization scripts available in MATLAB's optimization toolbox, and you can even make your own. Try to experiment with different scripts to see how it affects the results. Optimization scripts are often used to solve vehicle dynamics problems.

Analysis



Plotting

Plotting a tire model versus time does not provide much insight on the tire's performance. We will want to have a look at forces versus tire conditions. To plot the model using our script, you will need set '*6 - Visualisation, Polynomial Fits, and Lookup Tables*' as your working folder and open the script '*Main.m*'.

This script contains sections for each tire conditions and other sections to plot data only. The various tire conditions can be modified to view the performance window of interest. With our equations, we can observe how tire performance varies with camber, pressure, toe, slip angle, slip ratio, normal load.

Basic Results Review

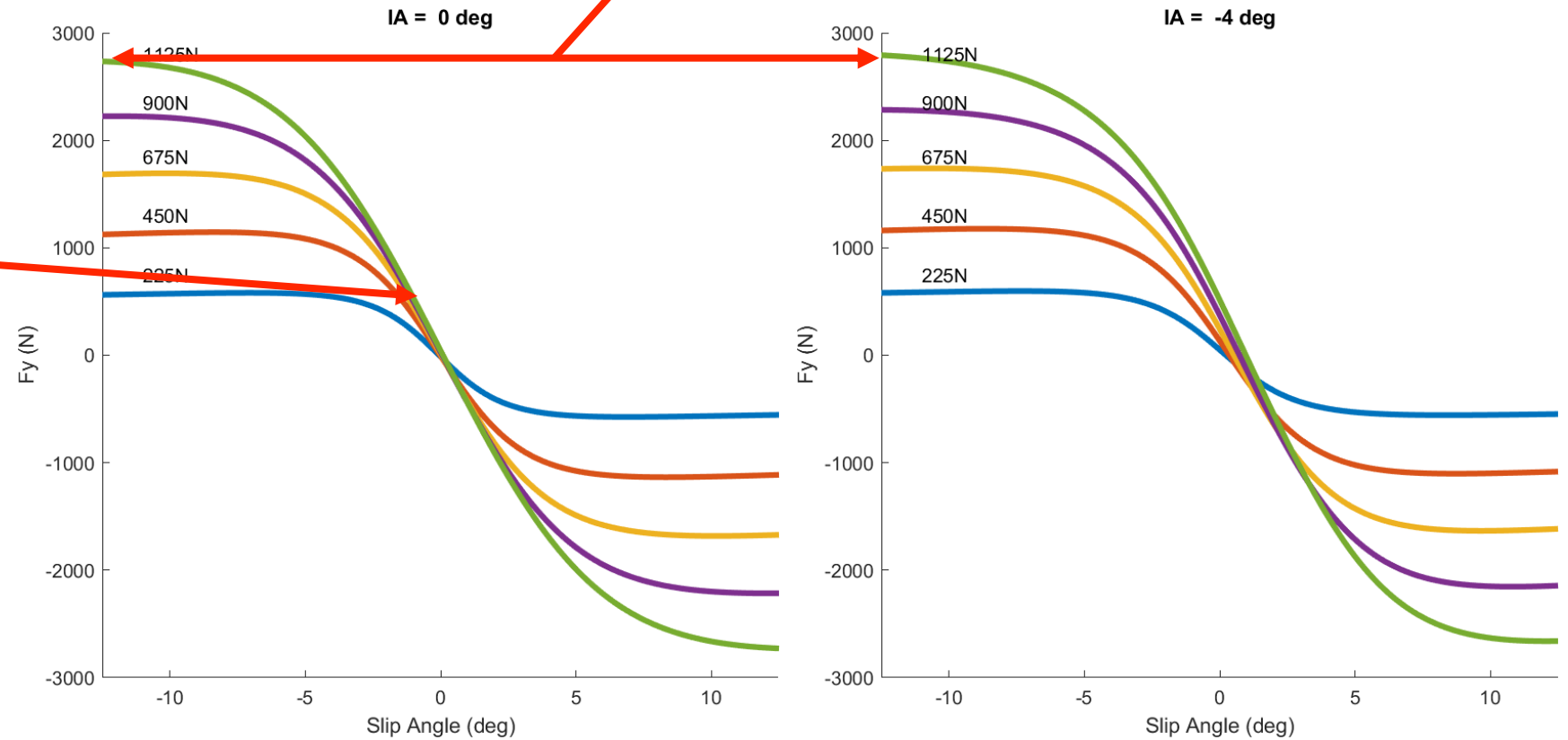
- There are two parts to reviewing results.
- 1. Comparing different tires to choose the one to run on the car
 - A good place to start is comparing temperature rise times. We run all events from cold tires, so we want to reach peak performance as quickly as possible.
 - Other great comparators are peak longitudinal and lateral forces.
 - Exploring all other aspects of the tires deepen insight and open new design opportunities.
- 2. Determining the performance window of the selected tire
 - What are the best operating slip ratios and slip angles?
 - What is the best combined window?
 - What camber angles and pressures to aim for?
 - Etc..

For both results review, it is important to review vehicle data and determine the actual use window of the tires before making a choice. The next slide quickly demonstrates interesting aspects of a lateral force versus slip angle plot. It is always great to refer to online resources since they can provide new ideas ([an example](#), [another example](#), [yet another example](#))!

Results Review Example

Compare peak lateral force produced at different camber angles

Observe the changes in lateral force at small slip angles. This region can represent the turn-in portion of a corner. A big variations in lateral force will mean that when large load transfers occur, less overall grip is generated.



[A great source to understand the grip circle.](#)

Final Remarks



Challenges

16-inch tires are not as straight forward as the larger tires. The 16-inch outer radius is too small for the TTC to properly install them on their rig. For this reason, they had to build an adapter. Unfortunately, using this adapter also means that the tire can only be evaluated under free rolling conditions. Drive/brake data allows us to fit the longitudinal coefficients and combined data allows us to fit all the combined slip coefficients. This leaves us with temperature sweeps, tire vertical spring rates, pure lateral performance and self-aligning moment in pure lateral conditions.

This brings up a few questions:

- How do we really know that the 16-inch tire is better than its 18-inch twin?
- Besides tire forces and moments, what are the gains of a smaller tire?
- If we pick a 16-inch tire, how do we decide between the 6.0" or 7.5" wide tires?

Suspension Kinematics

The vehicle suspension allows us to control the tires to ensure it is always in the optimal operating window. A thorough model helps set kinematics targets early in the design stage and really maximize grip generation.

Our current suspension design focuses on maximizing lateral grip through its kinematics curve. The reason? We just don't see as much gain for reaching peak longitudinal grip than we do for peak lateral grip. This stays a very limited approach. Tires are rarely (maybe even never if we forget about the acceleration event) in a pure lateral state. There is always a combination of slip angle and slip ratio calculations happening, and this is where we would ideally define our best kinematics curve. Tradeoffs in camber and toe gain will have a severe impact on our overall grip around the corner and driver feel!

Future Work

There are many opportunities of improvements for our modeling process.

- The data preprocessing requires rework to decrease time and improve robustness of the processed data.
 - The better the preprocessed data, the better the final model fit
- Model fitting optimization robustness improvements
 - The optimization is currently limited by past model fits. The script would require a more open approach to allow the parameters to be better narrowed down.
- A user-friendly GUI for fitting and analysis.
- Finding a better solution to missing 16-inch test data.